

Himalayan Expedition Data Analysis

```
members <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-09-22/members.csv')
```

members

```
# A tibble: 76,519 × 21
  expedition_id member_id peak_id peak_name year season sex age
  <chr>         <chr>    <chr>   <chr>    <dbl> <chr> <chr> <dbl>
1 AMAD78301     AMAD78301-01 AMAD    Ama Dablam 1978 Autumn M     40
2 AMAD78301     AMAD78301-02 AMAD    Ama Dablam 1978 Autumn M     41
3 AMAD78301     AMAD78301-03 AMAD    Ama Dablam 1978 Autumn M     27
4 AMAD78301     AMAD78301-04 AMAD    Ama Dablam 1978 Autumn M     40
5 AMAD78301     AMAD78301-05 AMAD    Ama Dablam 1978 Autumn M     34
6 AMAD78301     AMAD78301-06 AMAD    Ama Dablam 1978 Autumn M     25
7 AMAD78301     AMAD78301-07 AMAD    Ama Dablam 1978 Autumn M     41
8 AMAD78301     AMAD78301-08 AMAD    Ama Dablam 1978 Autumn M     29
9 AMAD79101     AMAD79101-03 AMAD    Ama Dablam 1979 Spring M     35
10 AMAD79101    AMAD79101-04 AMAD    Ama Dablam 1979 Spring M     37
# i 76,509 more rows
# i 13 more variables: citizenship <chr>, expedition_role <chr>, hired <lgl>,
#   highpoint_metres <dbl>, success <lgl>, solo <lgl>, oxygen_used <lgl>,
#   died <lgl>, death_cause <chr>, death_height_metres <dbl>, injured <lgl>,
#   injury_type <chr>, injury_height_metres <dbl>
```

Introduction

In this project, we are working with a subset of the Himalayan Database. The complete Himalayan Database is a compilation of records for all expeditioners that have climbed in the Himalayas, based on the expedition archives of journalist Elizabeth Hawley. The member dataset contains information of all individuals who participated in expeditions from 1905 to Spring 2019 to over 465 significant peaks in Nepal. More information about the dataset can be [found here \(Tidy Tuesday project\)](#) and [here \(Himalayan Database\)](#). The focus of this project is to demonstrate data wrangling, data analysis, and data visualization using R.

We will be using this dataset to answer the following questions:

Question 1: Which 10 countries had the most successful expedition members?

Question 2: Are there differences between the number of members who successfully and unsuccessfully completed expeditions considering the season in which the expeditions occurred?

Question 3: Has expedition success changed over time?

Analysis

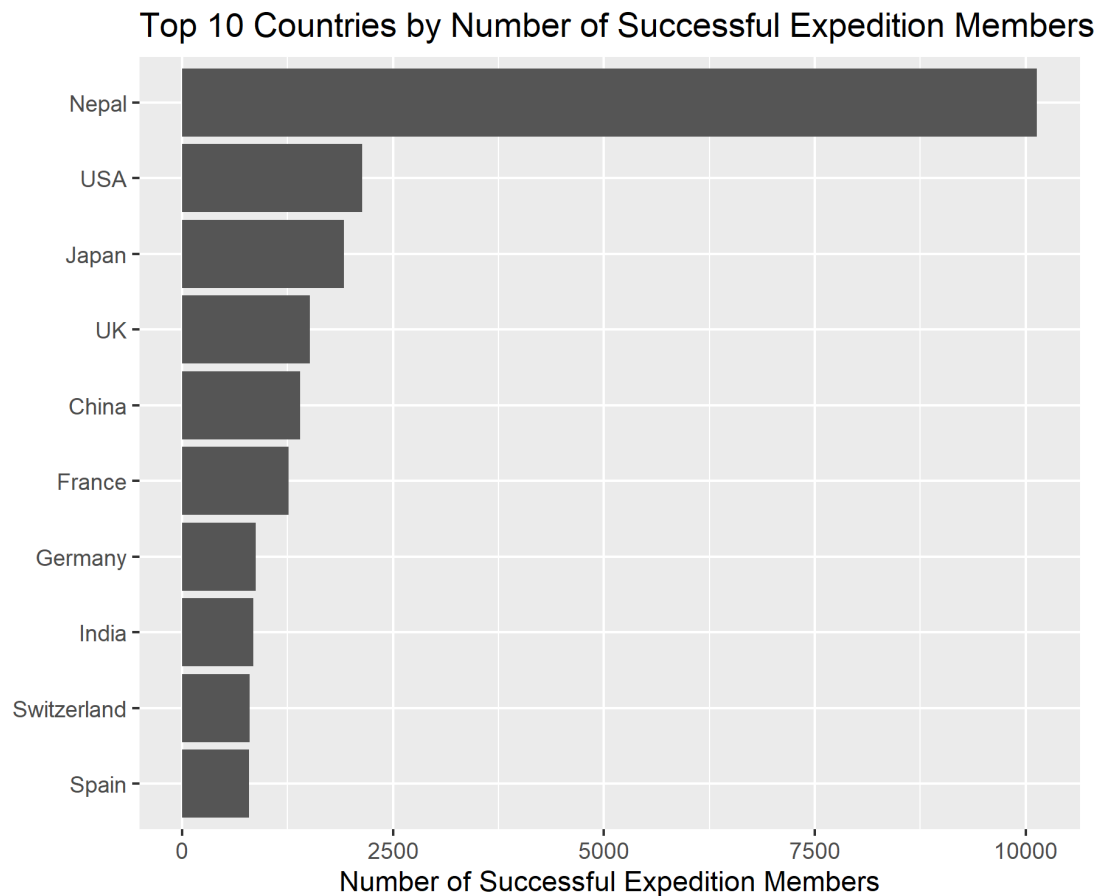
Question 1: Which 10 countries had the most successful expedition members?

```
# DATA WRANGLING
q1_data <- members |>
  filter(!is.na(citizenship), success == TRUE) |>
  count(citizenship, sort = TRUE) |>
  slice_head(n = 10)
```

q1_data

```
# A tibble: 10 × 2
  citizenship      n
  <chr>         <int>
1 Nepal         10134
2 USA            2131
3 Japan         1919
4 UK             1513
5 China          1401
6 France         1260
7 Germany         873
8 India           842
9 Switzerland     802
10 Spain          792
```

```
# PLOT
ggplot(q1_data, aes(y = reorder(citizenship, n), x = n)) +
  geom_col() +
  labs(title = "Top 10 Countries by Number of Successful Expedition Members")
+
  scale_y_discrete(name = NULL) +
  scale_x_continuous(name = "Number of Successful Expedition Members")
```



Key Findings: Nepal had substantially more successful expedition members than any other country in the dataset, followed by the United States and several other countries with comparatively high numbers of successful climbers. The distribution is heavily concentrated among the leading countries, with a substantial difference between the highest and lower-ranked countries.

Question 2: Are there differences between the number of members who successfully and unsuccessfully completed expeditions considering the season in which the expeditions occurred?

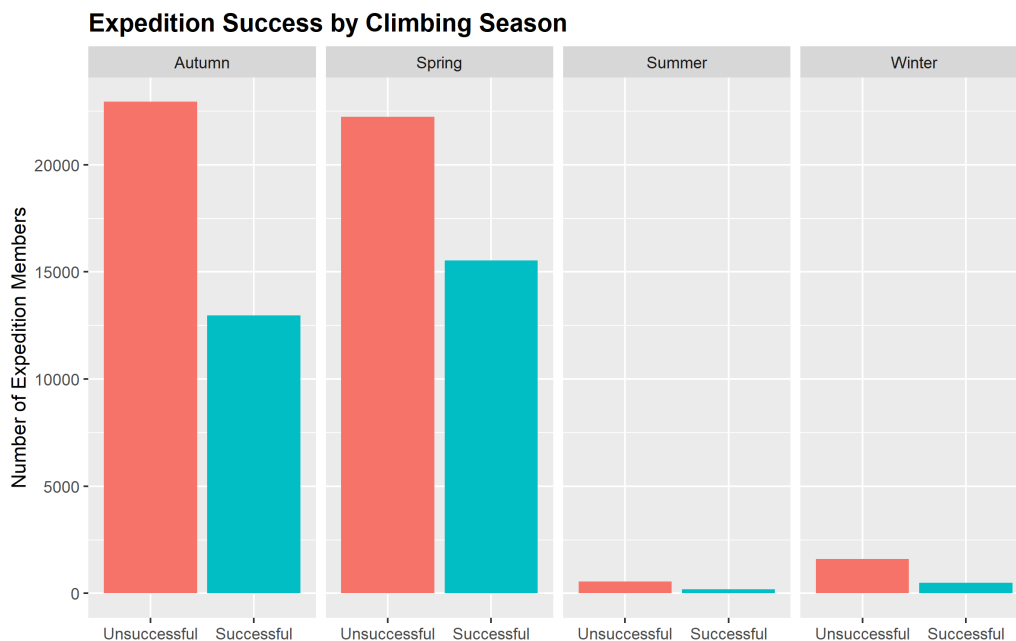
```
# DATA WRANGLING
q2_data <- members |>
  filter(season != "Unknown", !is.na(success)) |>
  group_by(season, success) |>
  summarize(count = n(), .groups = "drop")
```

```
q2_data
```

```
# A tibble: 8 × 3
  season success count
<chr>   <lgl>   <int>
1 Spring  TRUE     10000
2 Spring  FALSE     1000
3 Summer  TRUE     10000
4 Summer  FALSE     1000
5 Autumn  TRUE     10000
6 Autumn  FALSE     1000
7 Winter  TRUE     10000
8 Winter  FALSE     1000
```

1	Autumn	FALSE	22933
2	Autumn	TRUE	12962
3	Spring	FALSE	22243
4	Spring	TRUE	15539
5	Summer	FALSE	543
6	Summer	TRUE	197
7	Winter	FALSE	1601
8	Winter	TRUE	500

```
# PLOT
ggplot(q2_data, aes(x = success, y = count, fill = success)) +
  geom_col(position = "dodge") +
  scale_x_discrete(
    name = NULL,
    labels = c("FALSE" = "Unsuccessful", "TRUE" = "Successful")
  ) +
  scale_y_continuous(name = "Number of Expedition Members") +
  ggtitle("Expedition Success by Climbing Season") +
  scale_fill_discrete(guide = "none") +
  facet_wrap(~ season, nrow = 1) +
  theme(plot.title = element_text(size = 14, face = "bold"))
```



Key Findings: Spring and autumn accounted for the majority of expedition members, while summer and winter had substantially fewer. The proportion of successful members also varied by season, with spring and autumn showing higher success rates than summer and winter.

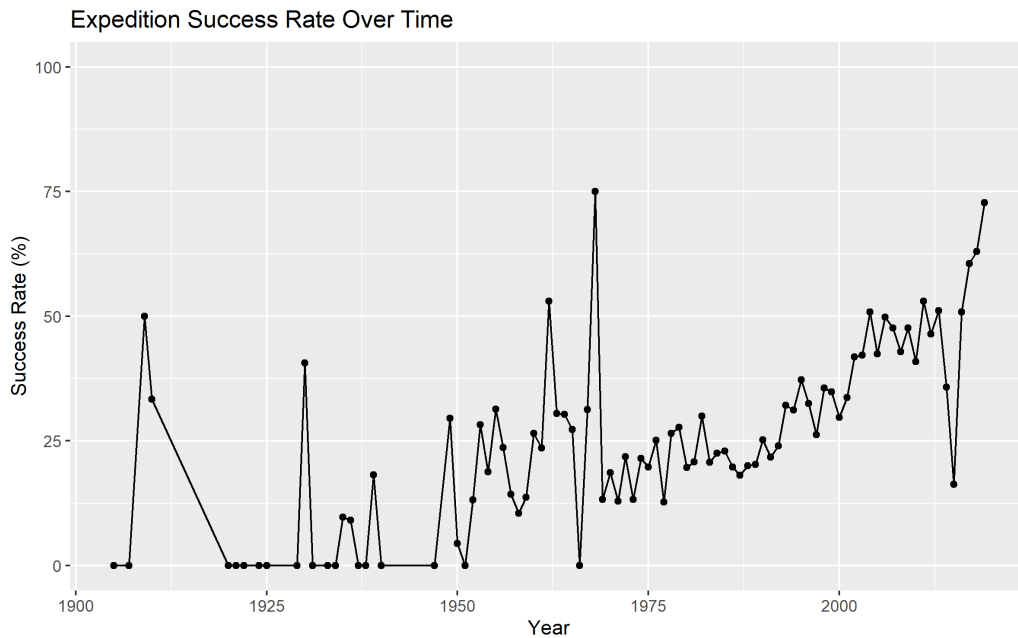
Question 3: Has expedition success changed over time?

```
# DATA WRANGLING
q3_data <- members |>
  filter(!is.na(year), !is.na(success)) |>
  group_by(year) |>
  summarize(
    total_members = n(),
    successful_members = sum(success),
    success_rate = successful_members / total_members * 100,
    .groups = "drop"
  )

q3_data
```

```
# A tibble: 92 × 4
   year total_members successful_members success_rate
  <dbl>         <int>          <int>         <dbl>
1  1905             9              0             0
2  1907             2              0             0
3  1909             2              1            50
4  1910             3              1           33.3
5  1920             5              0             0
6  1921            13              0             0
7  1922            22              0             0
8  1924            13              0             0
9  1925             2              0             0
10 1929            15              0             0
# i 82 more rows
```

```
# PLOT
ggplot(q3_data, aes(x = year, y = success_rate)) +
  geom_line() +
  geom_point() +
  scale_y_continuous(
    name = "Success Rate (%)",
    limits = c(0, 100)
  ) +
  scale_x_continuous(name = "Year") +
  ggtitle("Expedition Success Rate Over Time")
```



Key Findings: Expedition success rates generally increased over time, particularly from the 1970s onward. While earlier years show substantial fluctuations, the 2000s and 2010s generally experienced higher success rates than earlier decades. However, considerable year-to-year variation remains, and differences in the number of expedition members recorded in each year should be considered when interpreting these changes.

Conclusion

This analysis demonstrates how exploratory data analysis can be used to examine patterns within historical expedition data. By combining data cleaning, aggregation, and visualization, the project provides a clearer view of how expedition participation and outcomes vary across countries, seasons, and time. The analysis also highlights the importance of considering additional factors when interpreting expedition success, as the available data alone cannot explain the underlying reasons for these patterns.